

A counterexample to admissibility of the Graybill-Deal estimator

Reader's edition

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This is a typeset copy of the research draft. The mathematical claims have not been altered. Broken equation delimiters, missing spaces, and inconsistent estimator notation have been repaired. Short reader notes are included to make the proof easier to audit. The original source files remain unchanged.

Suggested reading order. First check Sections 1-3: these use standard normal theory, independence of sample means and variances, chi-square distributions, and the beta-gamma transformation. Next check the sign argument in Section 4 and the upper bound in Section 5. Those two sections contain the novel part of the proof. Section 6 then transfers the result to the bounded, clipped estimator. Section 7 is an extension and can be skipped on a first reading.

1. Result

Consider two independent samples of the same size n :

$$X_{ij} \sim N(\mu, \sigma_i^2), \quad i = 1, 2, \quad j = 1, \dots, n,$$

where $\mu \in \mathbb{R}$ and $\sigma_1^2, \sigma_2^2 > 0$ are unknown. Let S_i^2 be the usual unbiased sample variance and put

$$D = \bar{X}_2 - \bar{X}_1, \quad r = \frac{S_1^2}{S_1^2 + S_2^2}, \quad q = \frac{nD^2}{S_1^2 + S_2^2}.$$

The Graybill-Deal estimator is

$$\hat{\mu}_{\text{GD}} = \bar{X}_1 + Dr.$$

For $n = 13$, define

$$\tilde{w} = r + \frac{1}{2000}r(1-r)(1-2r)(4-q), \quad w_* = \min 1, \max 0, \tilde{w},$$

and

$$\hat{\mu}_* = \bar{X}_1 + Dw_*.$$

Then, under ordinary squared-error loss,

$$R(\mu, \sigma_1^2, \sigma_2^2; \hat{\mu}_*) < R(\mu, \sigma_1^2, \sigma_2^2; \hat{\mu}_{\text{GD}})$$

for every $\mu \in \mathbb{R}$ and every $\sigma_1^2, \sigma_2^2 > 0$.

The same construction, with a sample-size-dependent sufficiently small positive coefficient in place of $1/2000$, works for every equal sample size $n \geq 10$.

The key point is the dependence on q , and hence on D^2 . This places the competitor outside the class $\bar{X}_1 + D\phi(S_1^2, S_2^2)$ in which Duanmu, Roy, and Schritteser proved relative admissibility.

2. Exact risk reduction

Let

$$\lambda = \sigma_1^2 + \sigma_2^2, \quad \theta = \frac{\sigma_1^2}{\lambda}, \quad \nu = n - 1.$$

The known-variance estimator

$$T = \bar{X}_1 + \theta D$$

is independent of (D, S_1^2, S_2^2) . Therefore, for any data-dependent weight $w = w(D^2, S_1^2, S_2^2)$,

$$R(w) - R(r) = \mathbb{E}[D^2 \{(w - \theta)^2 - (r - \theta)^2\}]. \quad (1)$$

Reader note. Write an estimator with weight w as $T + D(w - \theta)$. Since $T - \mu$ has mean zero and is independent of the other factor, the cross-term vanishes after expanding the two squared errors. This is why w is allowed to depend on D^2 .

Put

$$V = \frac{nD^2}{\lambda} \sim \chi_1^2, \quad U_i = \frac{\nu S_i^2}{\sigma_i^2} \sim \chi_\nu^2,$$

with V, U_1, U_2 mutually independent. If

$$p(r) = r(1-r)(1-2r), \quad h = p(r)(4-q), \quad w = r + \varepsilon h,$$

then (1) becomes

$$\frac{n}{\lambda}R(w) - R(r) = 2\varepsilon B_\theta + \varepsilon^2 C_\theta, \quad (2)$$

where

$$B_\theta = \mathbb{E}[V(r - \theta)h], \quad C_\theta = \mathbb{E}[Vh^2]. \quad (3)$$

Thus it is enough to prove

$$B_\theta < 0 \quad \text{and} \quad 0 < \varepsilon < \inf_{0 < \theta < 1} \frac{-2B_\theta}{C_\theta}. \quad (4)$$

3. Beta reduction

It is convenient first to use

$$g = p(r) \left(1 - \frac{q}{4}\right), \quad h = 4g.$$

Let

$$P = \frac{U_1}{U_1 + U_2}, \quad L = U_1 + U_2.$$

Then

$$P \sim \text{Beta}(a, a), \quad L \sim \chi_{2\nu}^2, \quad a = \frac{\nu}{2},$$

and P, L, V are independent. If

$$d = \theta P + (1 - \theta)(1 - P),$$

then

$$r = \frac{\theta P}{d}, \quad q = \frac{\nu V}{Ld}.$$

Use

$$\mathbb{E}V = 1, \quad \mathbb{E}V^2 = 3, \quad \mathbb{E}V^3 = 15,$$

$$\mathbb{E}L^{-1} = \frac{1}{2(\nu - 1)}, \quad \mathbb{E}L^{-2} = \frac{1}{4(\nu - 1)(\nu - 2)}.$$

Writing $B_g = \mathbb{E}[V(r - \theta)g]$ and $C_g = \mathbb{E}[Vg^2]$ gives the exact one-dimensional formulas

$$B_g = \mathbb{E}_P \left[(r - \theta)p(r) \left\{ 1 - \frac{3\nu}{8(\nu - 1)d} \right\} \right], \quad (5)$$

$$C_g = \mathbb{E}_P \left[p(r)^2 \left\{ 1 - \frac{3\nu}{4(\nu - 1)d} + \frac{15\nu^2}{64(\nu - 1)(\nu - 2)d^2} \right\} \right]. \quad (6)$$

Reader note. These equations follow by conditioning on P and using $\mathbb{E}V = 1$, $\mathbb{E}V^2 = 3$, $\mathbb{E}V^3 = 15$, together with the two inverse moments of L . Checking those coefficients is a useful low-cost audit of the reduction.

4. Strict negativity of the linear term for $n = 13$

Set

$$s = 2\theta - 1, \quad x = 2P - 1.$$

The density of x is

$$K_a(1 - x^2)^{a-1}, \quad -1 < x < 1,$$

where $K_a > 0$, and

$$d = \frac{1 + sx}{2},$$

$$r - \theta = \frac{(1 - s^2)x}{2(1 + sx)},$$

$$p(r) = -\frac{(1 - s^2)(1 - x^2)(s + x)}{4(1 + sx)^3}.$$

For $n = 13$, $\nu = 12$, $a = 6$, and

$$\alpha = 1 - \frac{3\nu}{4(\nu - 1)} = \frac{2}{11}.$$

Equation (5) is

$$B_g = -\frac{K_a(1 - s^2)^2}{8}I(s), \tag{7}$$

where

$$I(s) = \int_{-1}^1 \frac{(1 - x^2)^6 x(s + x)(\alpha + sx)}{(1 + sx)^5} dx. \tag{8}$$

Let $z = s^2$ and

$$M_j = \int_0^1 x^{2j}(1 - x^2)^6 dx.$$

Pairing x and $-x$ in (8), followed by the absolutely convergent expansion of $(1 - s^2 x^2)^{-5}$, gives

$$I(s) = \sum_{m=0}^{\infty} M_{m+1} Q_m z^m. \tag{9}$$

The first coefficients are

$$Q_0 = \frac{4}{11}, \quad Q_1 = -\frac{116}{33}, \quad Q_2 = \frac{232}{11}.$$

For every $m \geq 3$,

$$(2m+1)Q_m = \frac{1}{11} \left[8232 + 20148 \binom{m-3}{1} + 23236 \binom{m-3}{2} + 13040 \binom{m-3}{3} + 2880 \binom{m-3}{4} \right] > 0. \quad (10)$$

Moreover,

$$\frac{M_2}{M_1} = \frac{3}{17}, \quad \frac{M_3}{M_1} = \frac{15}{323}.$$

Dropping the positive terms in (9) after $m = 2$ yields

$$\frac{I(s)}{M_1} \geq \frac{4}{11} - \frac{116}{187}z + \frac{3480}{3553}z^2. \quad (11)$$

The quadratic on the right has its minimum at $z = 19/60$, and that minimum is

$$\frac{1489}{5610} > 0.$$

Since

$$M_1 = \frac{1024}{45045},$$

(7) is strictly negative for every $-1 < s < 1$, equivalently every $0 < \theta < 1$.

Reader note. This is the central sign argument. Only Q_1 is negative. Retaining Q_0, Q_1, Q_2 and discarding the positive tail produces a quadratic lower bound whose completed-square minimum is positive.

5. A uniform bound for the quadratic term

Put

$$K_2 = \frac{\nu^2}{4(\nu-1)(\nu-2)}.$$

In (6), discard the negative middle term and use

$$(s+x)^2 \leq (1+sx)^2.$$

This gives

$$C_g \leq \frac{K_a(1-s^2)^2}{16}H, \quad (12)$$

where, for $c = 1/4$,

$$H = J_4 + 60c^2K_2J_6$$

and

$$J_k = \sup_{|s| \leq 1} \int_{-1}^1 \frac{(1-x^2)^{a+1}}{(1+sx)^k} dx.$$

Pairing x and $-x$ shows that the supremum is attained at $|s| = 1$. Hence

$$J_4 = 2^{2a-1} B(a+2, a-2), \quad J_6 = 2^{2a-3} B(a+2, a-4).$$

For $a = 6$,

$$K_2 = \frac{18}{55}, \quad J_4 = \frac{256}{165}, \quad J_6 = \frac{64}{9}, \quad H = \frac{1696}{165}.$$

Combining (7), (11), and (12),

$$\frac{-2B_g}{C_g} \geq \frac{95296}{40585545} \approx 0.002348028.$$

Since $h = 4g$,

$$B_\theta = 4B_g, \quad C_\theta = 16C_g,$$

so

$$\inf_{0 < \theta < 1} \frac{-2B_\theta}{C_\theta} \geq \frac{23824}{40585545} \approx 0.000587007. \quad (13)$$

The proposed value

$$\varepsilon = \frac{1}{2000} = 0.0005$$

is smaller than the bound in (13). Equation (2) is therefore strictly negative for every $0 < \theta < 1$.

6. Clipping, symmetry, and unbiasedness

The unclipped weight is unbounded as $q \rightarrow \infty$, although its risk is finite for $n = 13$. Projection onto $[0, 1]$ removes this cosmetic issue. Because $0 < \theta < 1$,

$$\text{clip}_{[0,1]}(w) - \theta^2 \leq (w - \theta)^2$$

pointwise. By (1), clipping can only improve the risk.

Under interchange of the two populations,

$$r \mapsto 1 - r, \quad q \mapsto q, \quad p(r) \mapsto -p(r),$$

so $\tilde{w} \mapsto 1 - \tilde{w}$; clipping preserves this identity. The estimator is therefore exchange symmetric. Its weight is an even function of D , so $Dw(D^2, S_1^2, S_2^2)$ has expectation zero by symmetry. Thus the competitor is also unbiased.

Reader note. Projection onto $[0, 1]$ does not use the unknown θ . The interval merely contains every possible value of θ , so the projection inequality is simultaneous over the entire parameter space. Strict improvement of the unclipped rule is therefore preserved.

7. Extension to all equal sample sizes $n \geq 10$

The same series proof works with $\nu = n - 1 \geq 9$ and $a = \nu/2$. Here

$$\alpha = \frac{\nu - 4}{4(\nu - 1)},$$

and the first three series coefficients are

$$Q_0 = \frac{\nu - 4}{2(\nu - 1)}, \quad Q_1 = -\frac{\nu^2 + 4\nu + 40}{6(\nu - 1)}, \quad Q_2 = \frac{5\nu^2 - 19\nu - 28}{2(\nu - 1)}.$$

For $m \geq 3$, write

$$(2m + 1)Q_m = \sum_{j=0}^4 D_j \binom{m-3}{j}.$$

After multiplication by $4(\nu - 1)$, the five D_j 's are

$$\begin{aligned} 308\nu^2 - 896\nu - 672, & \quad 712\nu^2 - 1774\nu - 648, \\ 798\nu^2 - 1804\nu - 320, & \quad 440\nu^2 - 928\nu - 64, \\ 96\nu^2 - 192\nu. \end{aligned}$$

All five are positive for $\nu \geq 8$, as is Q_2 . Thus Q_1 is the only negative coefficient.

Define

$$\begin{aligned} L_\nu(z) &= \frac{\nu - 4}{2(\nu - 1)} - \frac{\nu^2 + 4\nu + 40}{2(\nu - 1)(\nu + 5)}z \\ &\quad + \frac{15(5\nu^2 - 19\nu - 28)}{2(\nu - 1)(\nu + 5)(\nu + 7)}z^2. \end{aligned}$$

Direct polynomial algebra shows

$$\ell_\nu := \min_{0 \leq z \leq 1} L_\nu(z) > 0 \quad (\nu \geq 8).$$

For completeness, this last inequality is elementary. The discriminant of L_ν has the sign of

$$N(\nu) = \nu^5 - 285\nu^4 + 992\nu^3 + 9812\nu^2 - 17280\nu - 22400.$$

It is negative for $8 \leq \nu \leq 134$, so the quadratic is positive there. One quick check is $N(8) < 0$; for $9 \leq \nu \leq 134$, put $H(\nu) = \nu^3 - 285\nu^2 + 992\nu + 9812$. Then $H(9) < 0$, $H'(\nu) < 0$ on that interval, and $N(\nu)/\nu^2 < H(\nu)$.

For $\nu \geq 135$, the vertex lies to the right of 1. Indeed, the vertex condition reduces to $\nu^3 - 139\nu^2 + 638\nu + 1120 > 0$, which holds at 135 and is increasing thereafter. Finally,

$$L_\nu(1) = \frac{6(12\nu^2 - 61\nu - 140)}{2(\nu - 1)(\nu + 5)(\nu + 7)} > 0.$$

For $\nu > 8$, the quadratic-risk bound is finite. With

$$M_1 = \frac{1}{2}B\left(\frac{3}{2}, a + 1\right),$$

$$J_4 = 2^{2a-1}B(a + 2, a - 2), \quad J_6 = 2^{2a-3}B(a + 2, a - 4),$$

$$K_2 = \frac{\nu^2}{4(\nu - 1)(\nu - 2)}, \quad H_\nu = J_4 + \frac{15}{4}K_2J_6,$$

every

$$0 < \varepsilon < \frac{M_1 \ell_\nu}{H_\nu}$$

makes

$$w_\varepsilon = r + \varepsilon p(r)(4 - q)$$

strictly dominate the Graybill-Deal weight. Thus the construction proves inadmissibility for every equal sample size $n \geq 10$.

8. Relation to the literature

Duanmu, Roy, and Schritteser prove that the Graybill-Deal estimator is admissible in the class

$$\mathcal{C}_1 = \{\bar{X}_1 + D\phi(S_1^2, S_2^2) : 0 \leq \phi \leq 1\}.$$

The present competitor depends on D^2 through q , so it is outside $\mathcal{C}_1$. It therefore does not contradict their Corollary 5.3.

Sinha and Mouqadem's extended-admissibility result does not rule out a dominance gain that vanishes as a variance ratio approaches zero or infinity. Pal and Lim's result is second-order/asymptotic and likewise does not rule out the finite- n , higher-order improvement above.

Relevant sources

- H. Duanmu, D. M. Roy, and D. Schritteser, "Admissibility is Bayes optimality with infinitesimals," especially Section 5.1 and Corollary 5.3.
- B. K. Sinha and O. Mouqadem, "Estimation of the common mean of two univariate normal populations," *Communications in Statistics - Theory and Methods* 11 (1982), 1603-1614.
- N. Pal and W. K. Lim, "A note on second-order admissibility of the Graybill-Deal estimator of a common mean of several normal populations," *Journal of Statistical Planning and Inference* 63 (1997), 71-78.

9. Independent checks and next work

The accompanying script `verify_gd_counterexample.py` checks all exact fractions used in the $n = 13$ certificate and independently evaluates the original three-dimensional chi-square risk integral by generalized Gauss-Laguerre quadrature.

A separate numerical search found a much larger bounded correction already at $n = 4$:

$$w_4 = r + \frac{r(1-r)(2r-1)5q-3-24(2r-1)^2}{100(q+1)}.$$

High-order independent quadratures and Monte Carlo calculations give a negative risk difference over dense parameter grids, and the endpoint asymptotic is favorable:

$$\frac{n}{\lambda}R(w_4) - R(r) = -\frac{627}{3200}\theta^{3/2} + o(\theta^{3/2}) \quad (\theta \downarrow 0).$$

This $n = 4$ rule is not yet a theorem: it still needs a uniform analytic or interval-arithmetic certificate on the interior.

The immediate priorities are:

- independently typeset and referee the $n \geq 10$ proof;
- check every convention against the original literature;
- seek a shorter Stein/unbiased-risk derivation;
- certify the $n = 4$ candidate and then address $n = 2, \dots, 9$;
- extend the search to unequal sample sizes.

A practical audit checklist

For a reader with a graduate mathematical statistics and measure-theory background, the following division is useful:

1. **Normal-theory reduction:** verify independence of T from (D, S_1^2, S_2^2) and expand the two risks to obtain (1).
2. **Distributional coordinates:** derive the laws of V, U_1, U_2 , then the independent beta and chi-square variables (P, L) , and verify (5)-(6).
3. **Coordinate algebra:** check the three displayed formulas for $d, r - \theta, p(r)$ after setting $s = 2\theta - 1$ and $x = 2P - 1$.
4. **Novel sign certificate:** reproduce the first three Q_m coefficients, verify positivity of the tail in (10), and complete the square in (11).
5. **Quadratic bound:** check $(s+x)^2 \leq (1+sx)^2$, the monotonicity after pairing x with $-x$, and the two beta-integral values J_4, J_6 .
6. **Final decision-theory step:** compare $1/2000$ with (13), then check that clipping is a contraction about every $\theta \in (0, 1)$.

Completing items 1-6 independently verifies the fixed- $n = 13$ counterexample. The more elaborate argument in Section 7 is needed only for the extension to all equal sample sizes $n \geq 10$.